BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 095 · Year 10 Review and Year 11 Bridge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Name the IDE facility used to write and change source code.

Q2. Which description matches an AND gate symbol?
A. Triangle with output circle
B. Curved input; pointed output
C. Flat input; rounded output
D. A decision diamond

Q3. “Adding 5 to a ticket count always calculates £5 per ticket.” Correct this. Explain why.

Q4. (a) Convert denary 182 to two-digit hexadecimal. (b) Convert hexadecimal B6 to denary. Show your calculation.

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I Do · Read new knowledge

Key vocabulary

Command word — Instruction in an exam question, such as state, describe or explain.

Mark scheme — Guidance showing the knowledge and reasoning that earn marks.

Retrieval practice — Recalling knowledge from memory without looking at notes.

Trace table — Table recording variables and other values as an algorithm runs.

Specification point — A named section of the OCR course content used to target revision.

New learning

Know the two papers: Paper 1 is Computer Systems (sections 1.1–1.6); Paper 2 covers algorithms, programming, robustness, Boolean logic and languages (2.1–2.5). Each paper counts for half the GCSE.

Diagnose the missed mark: Look at the command word and the actual mark scheme. Ask whether the problem was missing knowledge, an incomplete explanation, a tracing error or a calculation mistake.

Improve the response: Write a technical point, then link it to the effect in the situation. 'RAM is fast' is vague; 'RAM holds running instructions and working data so the CPU can access them quickly' is a stronger explanation.

Make a retrieval plan: Choose two named specification points. Test without notes, compare with the mark scheme, correct mistakes, then answer a different question a few days later.

Worked example

A learner answers 'RAM is fast' to a question asking why RAM is used for currently running programs. Improve this into a full TWO-mark explanation. [2 marks]

1. 'Explain' requires a point and a relevant reason, not only a vague adjective.

2. RAM stores the instructions and data currently being used by the computer [1].

3. The CPU can access that working information quickly when executing the running program [1].

4. Both parts refer to the role of RAM. 'RAM is fast' alone would not secure both marks.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A learner needs a durable portable drive. Explain ONE reason an SSD may be preferable to an HDD. [2 marks]

Worked method: An SSD has no moving parts [1], so it is less likely to be damaged by knocks when carried [1].

Guided 2

Trace this algorithm: total = 2; total = total + 3; print(total). State the total after both assignments and the output. [3 marks]

Worked method: After first line total=2 [1]; after second line total=5 [1]; the output is 5 [1].

Guided 3

A message claims to be from a bank and demands an immediate password reset through an unfamiliar link. Identify the likely attack and explain one safe response. [3 marks]

Worked method: Phishing [1]; do not use the link [1] and contact the bank using a known official route to check the request [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the assignment that totals the three drive capacities.

```
array drivesGB = [64,128,256]
total = 0
for i = 0 to 2
    ...
next i
print(total)
```

write

Question 1

A pupil has written a program that stores numbers in an array. Explain why a trace table can help identify a logic error. [3 marks] Source: Original OCR J277-style practice question.

Question 2

A class uses a Wi-Fi network. Describe TWO actions the school can take to improve network security. [4 marks] Source: Original OCR J277-style practice question.

Question 3

A school gives a student an assessment with 'state' and 'explain' questions. Describe how the student should respond differently to these two command words. [3 marks] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

Writing the name of a feature is always enough to gain two marks for an 'explain' question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The editor.

Q2. C — Flat input; rounded output

Q3. Use total = tickets * 5; addition does not calculate £5 for each ticket.

Q4. (a) B6. (b) 11 × 16 + 6 = 182.

I Do / We Do pages · Guided practice

Q1. An SSD has no moving parts [1], so it is less likely to be damaged by knocks when carried [1].

Q2. After first line total=2 [1]; after second line total=5 [1]; the output is 5 [1].

Q3. Phishing [1]; do not use the link [1] and contact the bank using a known official route to check the request [1].

You Do pages · Code First

```
array drivesGB = [64,128,256]
total = 0
for i = 0 to 2
    total = total + drivesGB[i]
next i
print(total)
```

Expected result: 448

You Do pages · Independent questions

Q1. It records relevant variable or array values [1] as the algorithm executes step by step [1], so the point where a value first becomes incorrect can be found [1].

Q2. Use strong passwords or authentication [1] to reduce unauthorised access [1]. Encrypt the wireless connection [1] to protect data against interception [1].

Q3. 'State' generally requires a concise fact or answer [1]. 'Explain' needs a technical point [1] linked to a reason or effect [1].

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Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: